

CHAPTER 7

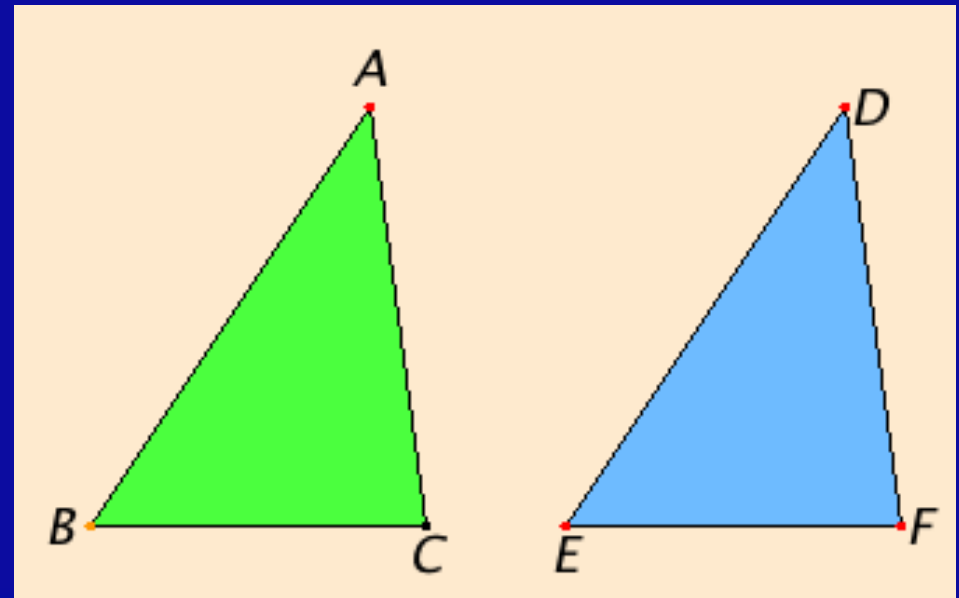
TRIANGLES



CHAPTER 7: TRIANGLES

Points:

- ❖ When three non-collinear points are joined together, a triangle is formed. It has 3 sides, 3 angles and 3 vertices. The sum of all angles of a triangle is 180°
- ❖ Two figures are said to be congruent if they are of same shape and of same size('congruent' means 'equal in all respects or figures whose shapes and sizes both are same'). Eg: 2 circles of the same radii.
- ❖ Two triangles are congruent, if sides and angles of a triangle are equal to the corresponding sides and angles of the other triangle. $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$, $AB = DE$, $BC = EF$, $AC = DF$ and if $\triangle ABC$ is congruent to $\triangle DEF$, then we write $\triangle ABC \cong \triangle DEF$



- ❖ In Congruent triangles, corresponding parts are equal and we write it in short, 'CPCT', i.e. Corresponding Parts of Congruent Triangles.

Axiom 7.1 [SAS (Side-Angle-Side) Congruence Rule]

Two triangles are congruent, if two sides and the included angle of a triangle are equal to the two sides and the included angle of the other triangle.

EXAMPLE 1 : In Fig. 7.8, $OA = OB$ and $OD = OC$. Show that
(i) $\triangle AOD \cong \triangle BOC$ and (ii) $AD \parallel BC$.

Given: $OA = OB$ and $OD = OC$

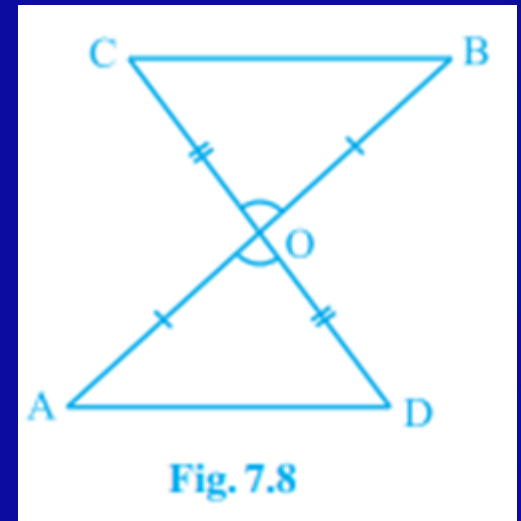
To prove that: (i) $\triangle AOD \cong \triangle BOC$ and (ii) $AD \parallel BC$

Proof: In $\triangle AOD$ and $\triangle BOC$,

$$OA = OB \text{ (given)}$$

$$OD = OC \text{ (given)}$$

$$\angle AOC = \angle BOC \text{ (Vertically Opposite Angles)}$$



$\therefore \triangle AOD \cong \triangle BOC$ (By SAS Congruence Rule)

Then by CPCT, $\angle A = \angle B$ (Alternate angles are equal)

$\Rightarrow AD \parallel BC$. HENCE PROVED

EXAMPLE 2: AB is a line segment and line l is its perpendicular bisector. If a point P lies on l , show that P is equidistant from A and B .

Given: $AC = BC$, $\angle ACP = \angle BCP = 90^\circ$

To prove that: $AP = BP$

Proof: In $\triangle ACP$ and $\triangle BCP$,

$AC = BC$ (given)

$\angle ACP = \angle BCP = 90^\circ$ (given)

$CP = CP$ (Common)

$\therefore \triangle ACP \cong \triangle BCP$ (By SAS Congruence Rule)

$\Rightarrow AP = BP$ (by CPCT)

i.e., P is equidistant from A and B .

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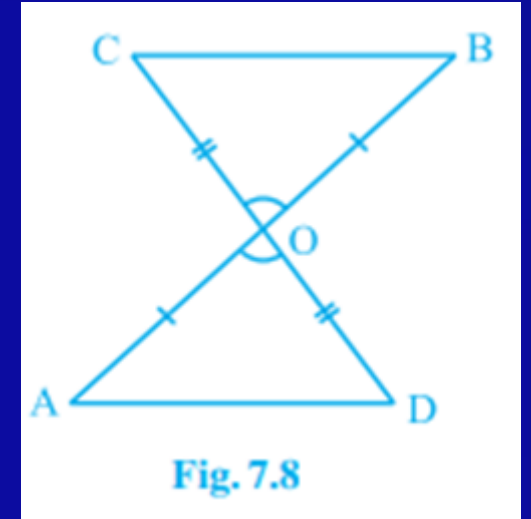


Fig. 7.8

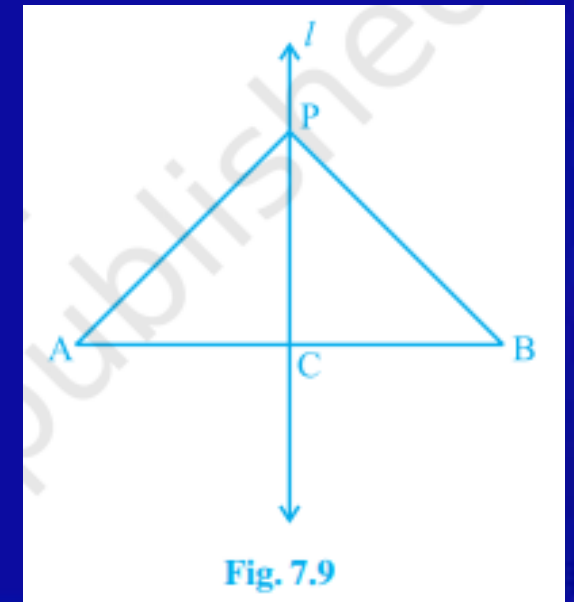


Fig. 7.9

THEOREM 7.1: (ASA congruence rule) : Two triangles are congruent if two angles and the included side of one triangle are equal to two angles and the included side of other triangle.

Given: ABC and DEF are the two triangles. $\angle B = \angle E$, $\angle C = \angle F$ and $BC = EF$

To Prove: $\triangle ABC \cong \triangle DEF$

Proof:

Case (i) : Let $AB = DE$

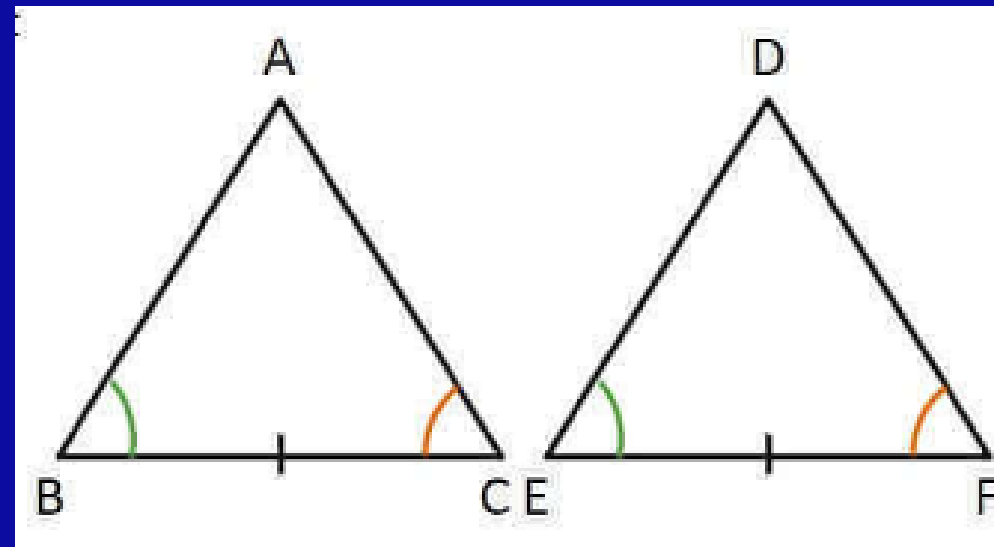
In $\triangle ABC$ & $\triangle DEF$,

$AB = DE$ (Assumed)

$\angle B = \angle E$ (Given)

$BC = EF$ (Given)

So, $\triangle ABC \cong \triangle DEF$ (By SAS rule)



Case (ii) : Let $AB > DE$

Let us assume a point P on AB such that $PB = DE$

Join PC.

In $\triangle PBC$ and $\triangle DEF$,

$$PB = DE \text{ (Assumed)}$$

$$\angle B = \angle E \text{ (Given)}$$

$$BC = EF \text{ (Given)}$$

$$\therefore \triangle PBC \cong \triangle DEF \text{ (by SAS rule) ----- ①}$$

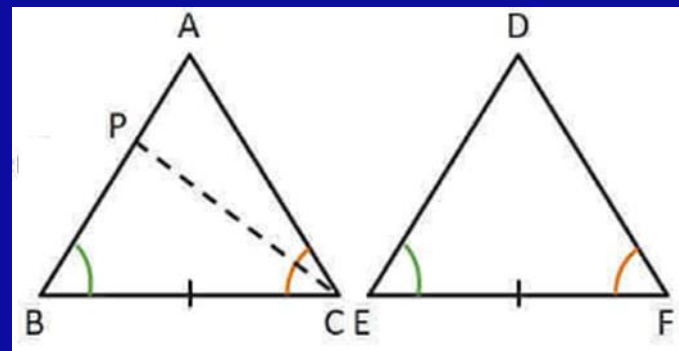
$$\Rightarrow \angle PCB = \angle DFE \text{ (by CPCT)}$$

$$\text{But, } \angle ACB = \angle DFE \text{ (Given)}$$

$$\Rightarrow \angle PCB = \angle ACB$$

This is possible only when P coincides with A.

Now ① becomes $\triangle ABC \cong \triangle DEF$



Case (iii) : Let $AB < DE$

Let us assume a point Q on DE such that $QE = AB$

Join QF.

In $\triangle ABC$ and $\triangle QEF$,

$AB = QE$ (Assumed)

$\angle B = \angle E$ (Given)

$BC = EF$ (Given)

$\therefore \triangle ABC \cong \triangle QEF$ (by SAS rule) ----- ②

$\Rightarrow \angle ACB = \angle QFE$ (by CPCT)

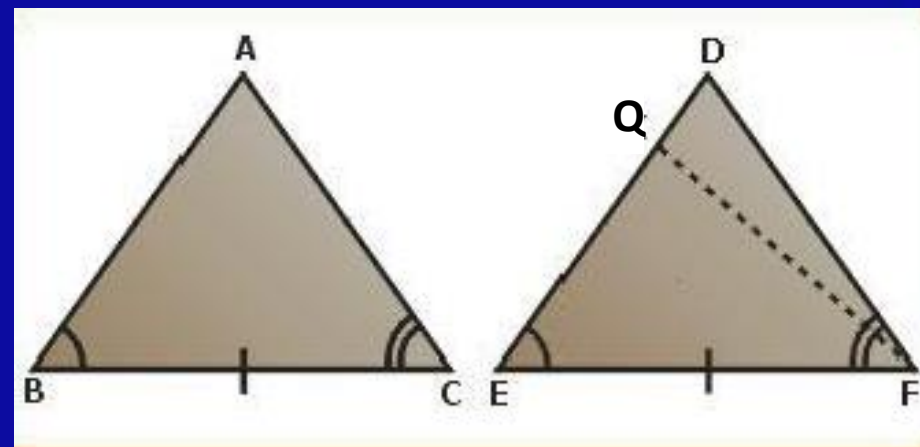
But, $\angle ACB = \angle DFE$ (Given)

$\Rightarrow \angle QFE = \angle DFE$

This is possible only when Q coincides with D.

Now ② becomes **$\triangle ABC \cong \triangle DEF$**

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EXAMPLE 2: Line-segment AB is parallel to another line-segment CD. O is the mid-point of AD (see Fig. 7.15).

Show that (i) $\triangle AOB \cong \triangle DOC$ (ii) O is also the mid-point of BC.

Given: $AB \parallel CD$, $OA = OD$

To prove: (i) $\triangle AOB \cong \triangle DOC$ (ii) O is also the mid-point of BC.

Proof:

Consider $\triangle AOB$ and $\triangle DOC$,

$\angle AOB = \angle DOC$ (Vertically opposite angles)

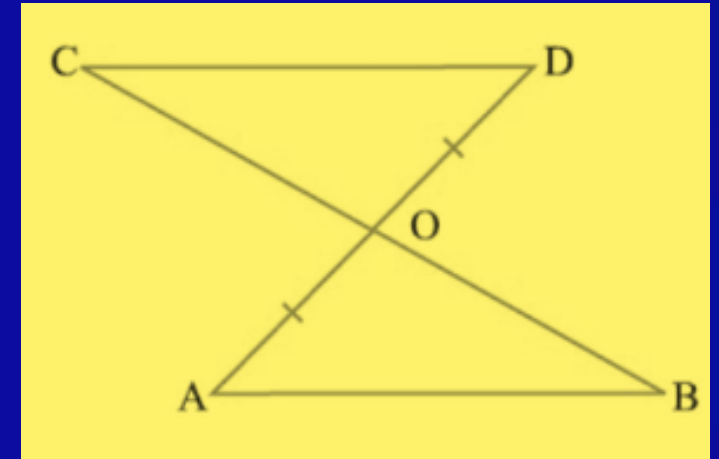
$OA = OD$ (Given)

$\angle A = \angle D$ (Alternate angles)

$\therefore \triangle AOB \cong \triangle DOC$ (ASA rule)

$\Rightarrow OB = OC$ (CPCT)

i.e., O is the mid-point of BC.



AAS (Angle-Angle-Side) Congruence Rule

Two triangles are congruent, if any two pairs of angles and one pair of corresponding sides are equal.

EXERCISE 7.1

1) In quadrilateral ACBD, $AC = AD$ and AB bisects $\angle A$. Show that $\triangle ABC \cong \triangle ABD$. What can you say about BC and BD?

Given: In quadrilateral ACBD, $AC = AD$ and AB bisects $\angle A$

To prove: $\triangle ABC \cong \triangle ABD$

Proof: In $\triangle ABC$ and $\triangle ABD$,

$$AC = AD$$

(Given)

$$\angle CAB = \angle DAB$$

(AB bisects $\angle A$)

$$AB = AB$$

(Common)

$$\therefore \triangle ABC \cong \triangle ABD$$

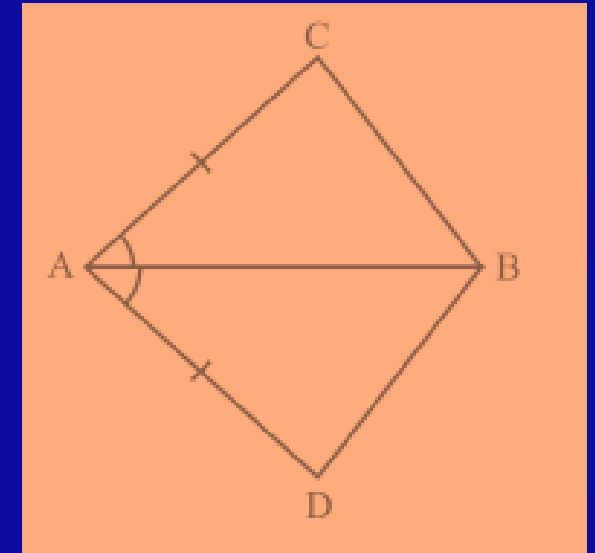
(By SAS congruence rule)

$$\Rightarrow BC = BD$$

(By CPCT)

Hence BC and BD are of equal lengths.

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2) ABCD is a quadrilateral in which $AD = BC$ and $\angle DAB = \angle CBA$. Prove that

(i) $\triangle ABD \cong \triangle BAC$

(ii) $BD = AC$

(iii) $\angle ABD = \angle BAC$.

Given: In quadrilateral ABCD, $AD = BC$ and $\angle DAB = \angle CBA$

To prove: (i) $\triangle ABD \cong \triangle BAC$

(ii) $BD = AC$

(iii) $\angle ABD = \angle BAC$

Proof:

In $\triangle ABD$ and $\triangle BAC$,

$AD = BC$ [Given]

$\angle DAB = \angle CBA$ [Given]

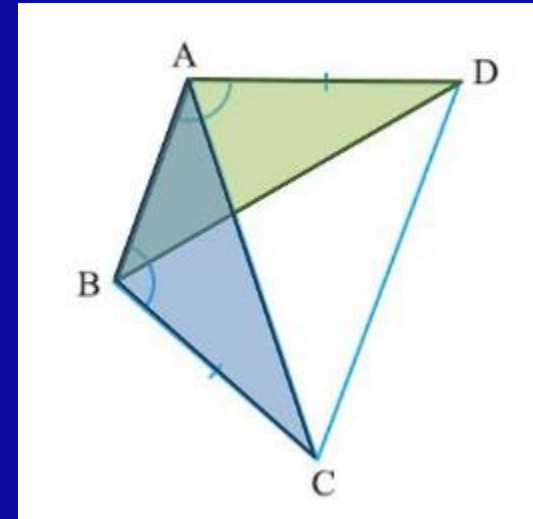
$AB = AB$ [Common side]

$\therefore \triangle ABD \cong \triangle BAC$ [SAS Congruence rule]

$\Rightarrow BD = AC$ [CPCT]

$\Rightarrow \angle ABD = \angle BAC$ [CPCT]

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3) AD and BC are equal perpendiculars to a line segment AB. Show that CD bisects AB.

Given: $AD \perp AB$, $BC \perp AB$ and $AD = BC$

To prove: CD bisects AB i.e., $OA = OB$

Proof:

In $\triangle OAD$ & $\triangle OBC$

$AD = BC$ (Given)

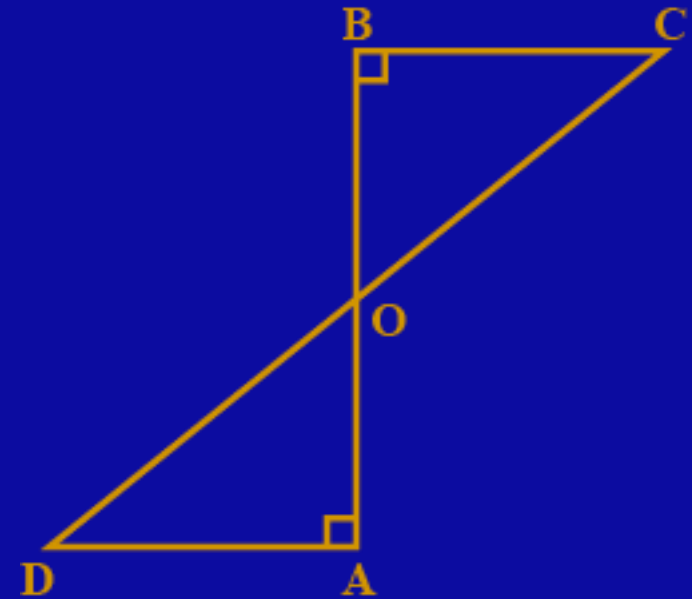
$\angle B = \angle A$ (90° each)

$\angle AOD = \angle BOC$ (Vertically Opposite Angles)

$\therefore \triangle AOD \cong \triangle BOC$ (AAS congruence rule)

$\Rightarrow OA = OB$ [CPCT]

Thus CD bisects AB



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4) l and m are two parallel lines intersected by another pair of parallel lines p and q . Show that $\triangle ABC \cong \triangle CDA$.

Given: $l \parallel m$ and $p \parallel q$

To prove: $\triangle ABC \cong \triangle CDA$

Proof:

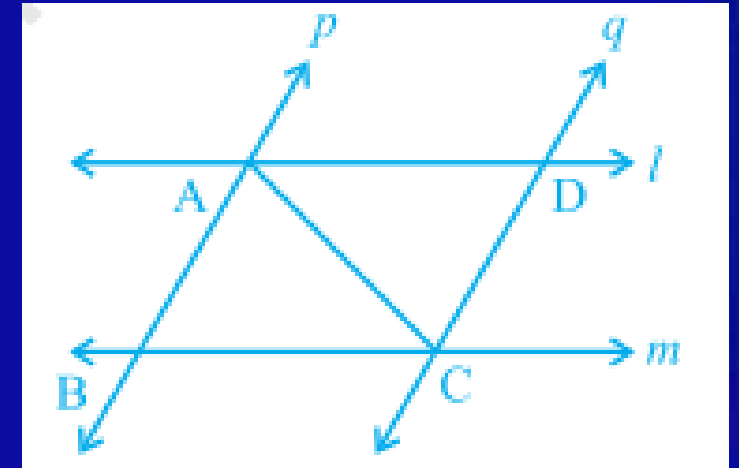
In $\triangle ABC$ and $\triangle CDA$,

$\angle BAC = \angle DCA$ (Alternate interior angles, as $p \parallel q$)

$AC = CA$ (Common)

$\angle BCA = \angle DAC$ (Alternate interior angles, as $l \parallel m$)

$\therefore \triangle ABC \cong \triangle CDA$ (By ASA congruence rule)



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5) Line l is the bisector of an angle $\angle A$ and B is any point on l . BP and BQ are perpendiculars from B to the arms of $\angle A$. Show that:

(i) $\triangle APB \cong \triangle AQB$

(ii) $BP = BQ$ or B is equidistant from the arms of $\angle A$.

Given: l is the bisector of $\angle A$ i.e., $\angle PAB = \angle QAB$
 $\angle P = \angle Q = 90^\circ$

To prove: i) $\triangle APB \cong \triangle AQB$ ii) $BP = BQ$

Proof:

In $\triangle APB$ and $\triangle AQB$

$\angle APB = \angle AQB$ (Each 90°)

$\angle PAB = \angle QAB$ (l is the angle bisector of $\angle A$)

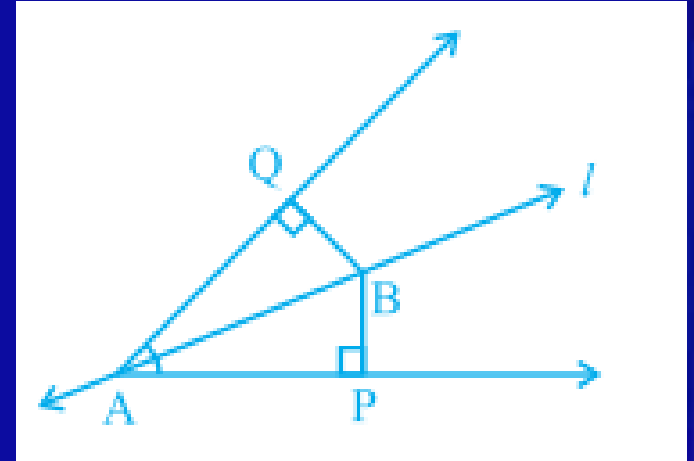
$AB = AB$ (Common)

$\therefore \triangle APB \cong \triangle AQB$ (By AAS congruence rule)

$\Rightarrow BP = BQ$ (By CPCT)

Hence B is equidistant from the arms of $\angle A$.

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6) In Fig. 7.21, $AC = AE$, $AB = AD$ and $\angle BAD = \angle EAC$. Show that $BC = DE$.

Given: $AC = AE$, $AB = AD$ and $\angle BAD = \angle EAC$

To prove: $BC = DE$

Proof: In $\triangle ABC$ and $\triangle ADE$,

$AC = AE$ (Given)

$AB = AD$ (Given)

Since, $\angle BAD = \angle EAC$

On adding $\angle DAC$ to both sides,

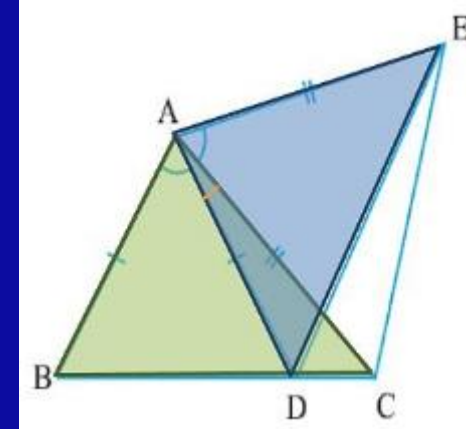
$\angle BAD + \angle DAC = \angle EAC + \angle DAC$

$\Rightarrow \angle BAC = \angle EAD$

Hence, $\triangle ABC \cong \triangle ADE$ (by SAS congruence rule)

$\Rightarrow BC = DE$ (by CPCT)

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7) AB is a line segment and P is its mid-point. D and E are points on the same side of AB such that $\angle BAD = \angle ABE$ and $\angle EPA = \angle DPB$. Show that

(i) $\triangle DAP \cong \triangle EBP$ (ii) $AD = BE$

Given: P is midpoint of AB i.e., $AP = BP$

$\angle BAD = \angle ABE$ and $\angle EPA = \angle DPB$

To Prove: (i) $\triangle DAP \cong \triangle EBP$ (ii) $AD = BE$

Proof: In $\triangle EBP$ and $\triangle DAP$,

$BP = AP$ (Given)

$\angle EBP = \angle DAP$ (Given)

Since, $\angle EPA = \angle DPB$

On adding $\angle EPD$ to both sides,

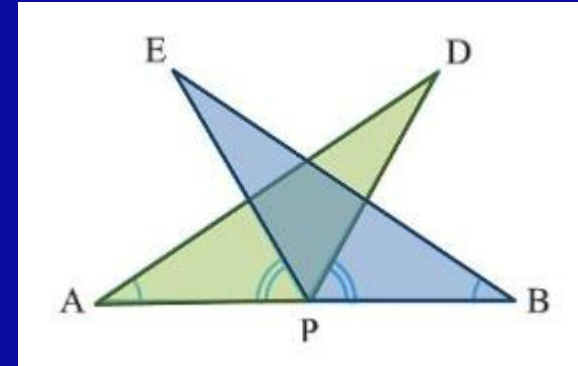
$\angle EPA + \angle EPD = \angle DPB + \angle EPD$

$\angle APD = \angle EPB$

$\therefore \triangle EBP \cong \triangle DAP$ (by ASA congruence rule)

$\Rightarrow AD = BE$ (using CPCT)

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8) In right triangle ABC, right angled at C, M is the mid-point of hypotenuse AB. C is joined to M and produced to a point D such that $DM = CM$. Point D is joined to point B. Show that:

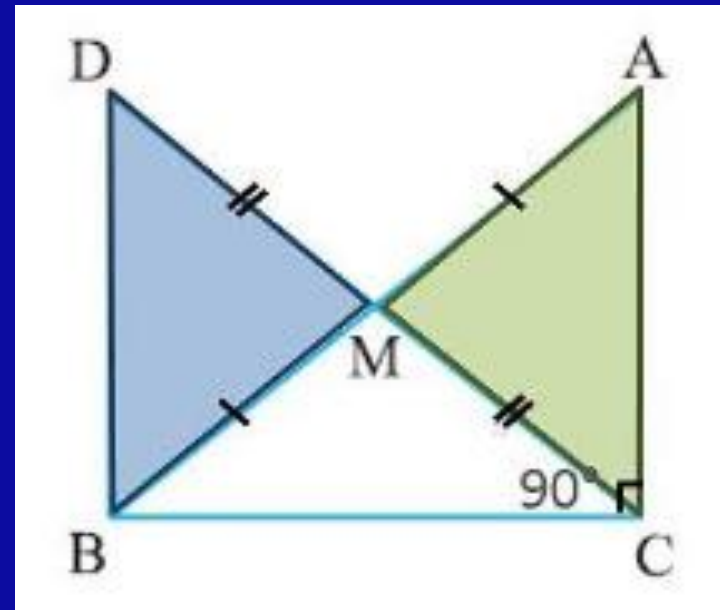
(i) $\triangle AMC \cong \triangle BMD$ (ii) $\angle DBC$ is a right angle.
(iii) $\triangle DBC \cong \triangle ACB$ (iv) $CM = \frac{1}{2} AB$

Given:

In right angled $\triangle ABC$, $\angle C = 90^\circ$,
 $AM = MB$ & $DM = CM$.

To show that:

- (i) $\triangle AMC \cong \triangle BMD$
- (ii) $\angle DBC$ is a right angle.
- (iii) $\triangle DBC \cong \triangle ACB$
- (iv) $CM = \frac{1}{2} AB$



Proof:

(i) In $\triangle AMC$ & $\triangle BMD$,

$$AM = BM$$

(Given)

$$\angle CMA = \angle DMB$$

(Vertically opposite angles)

$$CM = DM$$

(Given)

$$\therefore \triangle AMC \cong \triangle BMD$$

(by SAS congruence rule)

$$\Rightarrow AC = DB$$

(by CPCT)

$$\Rightarrow \angle ACM = \angle BDM$$

(by CPCT)(which are alternate angles)

Hence, $AC \parallel BD$ as alternate interior angles are equal.

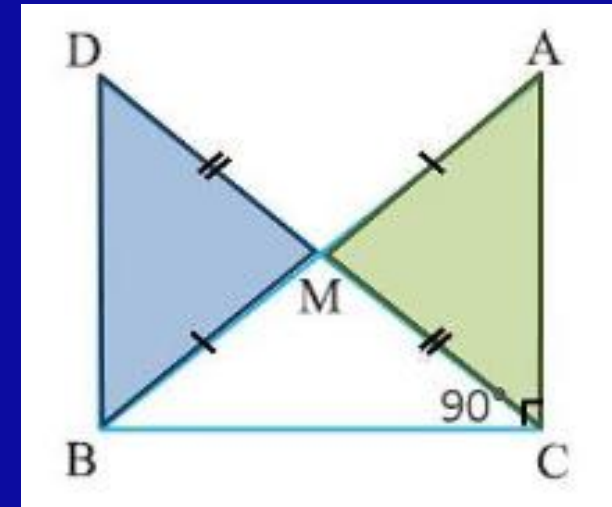
(ii) $\angle ACB + \angle DBC = 180^\circ$ (co-interiors angles)

$$\Rightarrow 90^\circ + \angle DBC = 180^\circ$$

$$\Rightarrow \angle DBC = 90^\circ$$

$$\text{Hence, } \angle DBC = 90^\circ$$

(Right Angle)



(iii) In $\triangle DBC$ & $\triangle ACB$,

$$BC = BC$$

(Common)

$$\angle ACB = \angle DBC$$

(90°)

$$DB = AC$$

(Proved)

$$\therefore \triangle DBC \cong \triangle ACB$$

(by SAS congruence rule)

(iv)

$$\Rightarrow AB = DC$$

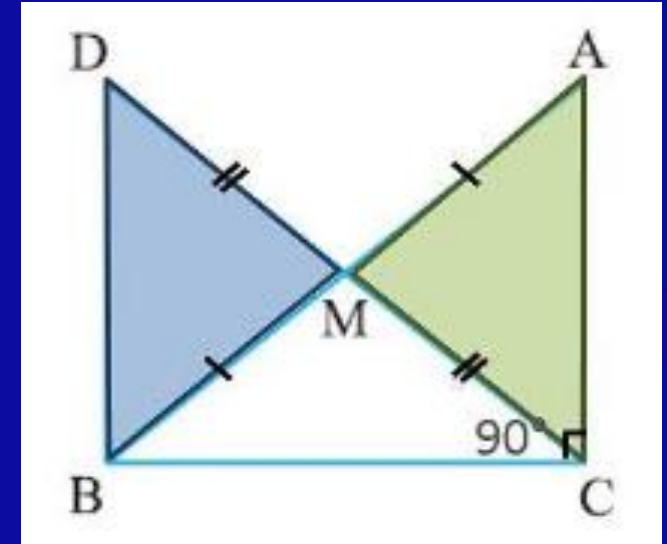
(By CPCT)

$$\Rightarrow AB = 2CM$$

($DM = CM$)

$$\Rightarrow \frac{1}{2} AB = CM$$

$$\therefore CM = \frac{1}{2} AB$$



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